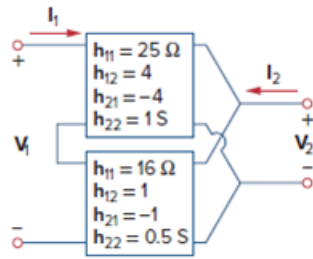


Problem

A series-parallel connection of two two-ports is shown in Fig. 19.119. Determine the z parameter representation of the network.

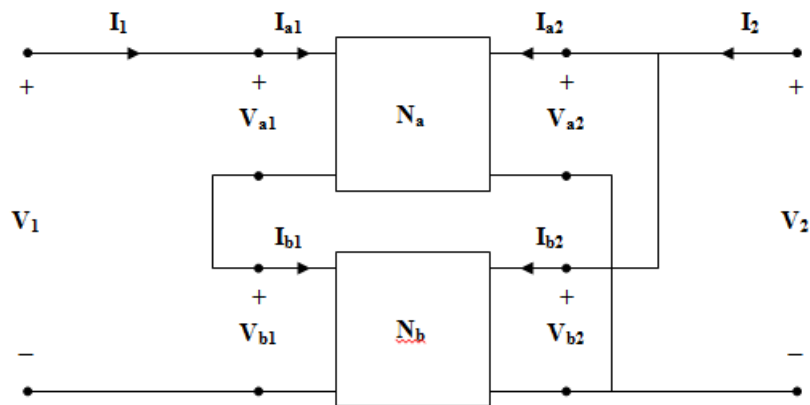
Figure 19.119



Step-by-step solution

Step 1 of 1

Consider the network shown below.



$$V_{a1} = 25 I_{a1} + 4 V_{a2} \quad (1)$$

$$I_{a2} = -4 I_{a1} + V_{a2} \quad (2)$$

$$V_{b1} = 16 I_{b1} + V_{b2} \quad (3)$$

$$I_{b2} = -I_{b1} + 0.5 V_{b2} \quad (4)$$

$$V_1 = V_{a1} + V_{b1}$$

$$V_2 = V_{a2} = V_{b2}$$

$$I_2 = I_{a2} + I_{b2}$$

$$I_1 = I_{a1}$$

Now, rewrite (1) to (4) in terms of I_1 and V_2

$$V_{a1} = 25 I_1 + 4 V_2 \quad (5)$$

$$I_{a2} = -4 I_1 + V_2 \quad (6)$$

$$V_{b1} = 16 I_{b1} + V_2 \quad (7)$$

$$I_{b2} = -I_{b1} + 0.5 V_2 \quad (8)$$

Adding (5) and (7),

$$V_1 = 25 I_1 + 16 I_{b1} + 5 V_2 \quad (9)$$

Adding (6) and (8),

$$I_2 = -4 I_1 - I_{b1} + 1.5 V_2 \quad (10)$$

$$I_1 = I_1 \quad (11)$$

$$\mathbf{I}_{b1} = \mathbf{I}_{a1} + \mathbf{I}_1 \quad (11)$$

Because the two networks $\mathbf{N}_a$ and $\mathbf{N}_b$ are independent,

$$\mathbf{I}_2 = -5\mathbf{I}_1 + 1.5\mathbf{V}_2$$

$$\text{or} \quad \mathbf{V}_2 = 3.333\mathbf{I}_1 + 0.6667\mathbf{I}_2 \quad (12)$$

Substituting (11) and (12) into (9),

$$\mathbf{V}_1 = 41\mathbf{I}_1 + \frac{25}{1.5}\mathbf{I}_1 + \frac{5}{1.5}\mathbf{I}_2$$

$$\mathbf{V}_1 = 57.67\mathbf{I}_1 + 3.333\mathbf{I}_2 \quad (13)$$

Comparing (12) and (13) with the following equations

$$\mathbf{V}_1 = z_{11}\mathbf{I}_1 + z_{12}\mathbf{I}_2$$

$$\mathbf{V}_2 = z_{21}\mathbf{I}_1 + z_{22}\mathbf{I}_2$$

indicates that

$$[\mathbf{z}] = \begin{bmatrix} 57.67 & 3.333 \\ 3.333 & 0.6667 \end{bmatrix} \Omega$$

Alternatively,

$$[\mathbf{h}_a] = \begin{bmatrix} 25 & 4 \\ -4 & 1 \end{bmatrix}, \quad [\mathbf{h}_b] = \begin{bmatrix} 16 & 1 \\ -1 & 0.5 \end{bmatrix}$$

$$[\mathbf{h}] = [\mathbf{h}_a] + [\mathbf{h}_b] = \begin{bmatrix} 41 & 5 \\ -5 & 1.5 \end{bmatrix} \quad \Delta_h = 61.5 + 25 = 86.5$$

$$[\mathbf{z}] = \begin{bmatrix} \frac{\Delta_h}{h_{22}} & \frac{h_{12}}{h_{22}} \\ \frac{-h_{21}}{h_{22}} & \frac{1}{h_{22}} \end{bmatrix} = \underline{\underline{\begin{bmatrix} 57.67 & 3.333 \\ 3.333 & 0.6667 \end{bmatrix} \Omega}}$$

as obtained previously.